

Lecture 4: Probability & Information Theory

BINF 4002

Outline

- 1 A quick note about expectation
- 2 Joint random variables, continued (Bayes rule, law of total probability)
- 3 Bayesian Statistics
- 4 Information Theory and Geometry
- 5 Statistics

Expectation and the Law of Large Numbers

Expectation is useful in statistics because the probability that an average of observed independent samples will converge to the true expected value approaches 1.

Law of Large Numbers:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} \mathbb{E}[X] \quad \text{as } n \rightarrow \infty$$

Given a Joint Distribution, What is a Marginal?

- Joint: $P(Y, X)$
- Marginal: $P(X) = ?$

[Visualization: 2D contour plot showing joint distribution with marginals projected on axes]

Why? Because Unions Sum!

Formally, a probability triple (X, Σ, P) is a tuple of three things:

- ① A set of possible samples the random variable can take: X
- ② The set of possible “events” that you are allowed to consider over these samples: $\Sigma \subseteq 2^X$
- ③ A measure over these samples and events that is a valid probability measure:
 $P : \Sigma \rightarrow [0, 1]$
 - $P(X) = 1$
 - For disjoint $A_i \in \Sigma$: $P(\bigcup_i A_i) = \sum_i P(A_i)$

Marginal is the Integral Over the Joint!

- Joint: $P(Y, X)$
- Marginal:

$$P(X) = \int_Y P(Y, X) dy$$

And the Conditional?

- Joint: $P(Y, X)$
- Marginal: $P(X) = \int_Y P(Y, X) dy$
- Conditional: $P(Y | X) = ?$

Conditional is the Joint, Renormalized

- Joint: $P(Y, X)$
- Marginal: $P(X) = \int_Y P(Y, X) dy$
- Conditional:

$$P(Y | X) = \frac{P(Y, X)}{P(X)}$$

Conditional is the Joint, Renormalized

- Joint: $P(Y, X)$
- Marginal: $P(X) = \int_Y P(Y, X) dy$
- Conditional:

$$P(Y | X) = \frac{P(Y, X)}{P(X)} = \frac{P(Y, X)}{\int_Y P(Y, X) dy}$$

Independent vs. Dependent Random Variables

If X, Y are independent, then:

$$P(X, Y) = P(X) \cdot P(Y)$$

and

$$P(X | Y) = P(X)$$

$$P(Y | X) \cdot P(X) = P(X | Y) \cdot P(Y)$$

$$P(Y | X) \cdot P(X) = P(X | Y) \cdot P(Y)$$

Equivalently...

$$P(Y | X) = \frac{P(X | Y) \cdot P(Y)}{P(X)} = \frac{P(X | Y) \cdot P(Y)}{\int_Y P(Y, X) dy}$$

$$P(Y | X) = \frac{P(X | Y) \cdot P(Y)}{P(X)}$$

$$\text{Posterior} = \frac{\text{Prior} \times \text{Likelihood}}{\text{Evidence}}$$

Key Points about Bayesian Inference!

A prior is **not** your “belief” or “guess”.

It is the marginal for the joint random variable.

Bayesian Inference is only valid if your prior is valid...

But if you use the right prior, it may be useful regardless.

*“All models are wrong
but some are useful.”*

— George E. P. Box

Information Theory

Information Axioms (Shannon)

- An event with probability 100% is perfectly unsurprising and yields no information.
- The less probable an event is, the more surprising it is and the more information it yields.
- If two independent events are measured separately, the total amount of information is the sum of the self-informations of the individual events.

The Unique Information Content Function

$$I(x) = -\log P(x)$$

Does this really quantify “information”?

Entropy: Expected Information Content!

$$H(X) = \mathbb{E}_X[I(X)]$$

$$H(X) = - \sum_x P(x) \log P(x)$$

Entropy is disorder/uncertainty...

but it's the average information content?

The average number of bits needed to encode a stream of i.i.d. random variables drawn from X is the average information content of X — a.k.a.
 $H(X)$

Coding Theorem Proof Argument (in 1 Sentence)

Basically every sequence that you'd actually ever see of n i.i.d. random variables from X lives in a set no bigger than $2^{nH(X)}$ — regardless of the values X can take on!

Coding Theorem Proof (in More Sentences)

- By the law of large numbers, the average observed total information content of a sampled sequence will converge to the expectation of the information content of the distribution (a.k.a., the entropy).
- This means that almost all observed sequences will have probability not far from $2^{-n \cdot H(X)}$.
- How many sequences make up this set of “almost all observed sequences”? It can't be more than (roughly) $1/2^{-n \cdot H(X)}$
 - Because otherwise the total probability of all those sequences together would be more than 1!
- So, almost all observed sequences will come from a set of size no bigger than roughly $2^{n \cdot H(X)}$

Coding Theorem: Formal Statement

Proof of Achievability. Fix some $\varepsilon > 0$, and let

$$p(x_1, \dots, x_n) = \Pr[X_1 = x_1, \dots, X_n = x_n]$$

The **typical set**, A_n^ε , is defined as:

$$A_n^\varepsilon = \left\{ (x_1, \dots, x_n) : \left| -\frac{1}{n} \log p(x_1, \dots, x_n) - H_n(X) \right| < \varepsilon \right\}$$

By the asymptotic equipartition property (AEP):

$$2^{-n(H(X)+\varepsilon)} \leq p(x_1, \dots, x_n) \leq 2^{-n(H(X)-\varepsilon)}$$

Therefore: $|A_n^\varepsilon| \leq 2^{n(H(X)+\varepsilon)}$, so $n(H(X) + \varepsilon)$ bits suffice.

Information content quantifies “information” because the law of large numbers and properties of logs mean we can translate between log probabilities and the log of the number of possible observed options across i.i.d. samples.

Independent Random Variables and Information

If X, Y are independent, then:

$$P(X, Y) = P(X) \cdot P(Y)$$

Taking logs:

$$\log(P(X) \cdot P(Y)) = \log(P(X)) + \log(P(Y))$$

Therefore: $I(X, Y) = I(X) + I(Y)$ for independent events.

What if I encode samples from a distribution P with a model from distribution Q ?

$$H(P, Q) = \mathbb{E}_P[I_Q] = - \sum_x P(x) \log Q(x)$$

Properties:

- $H(P, P) = H(P)$
- If $P \neq Q$, then $H(P, Q) > H(P)$

Caveat: What's information for continuous distributions? What's entropy?

Continuous Information and Entropy

- For a continuous random variable, $P(x)$ for any $x \in \mathcal{X}$ is 0.
- So, information content is infinite?
- ... What if we use $p_X(x)$ (the density)!
- $H(X) = \int_{\mathcal{X}} -\log(p_X(x)) dx$
- ... but what about units?

If each atom of X has size dx :

$$\int_S \frac{dP(x)}{dx} dx = P(S)$$

The density has units of “change in probability” / “size of X ”. What is the “log” of these units?

This is Not a Philosophical Problem!

- Suppose I have a uniform distribution from a to b
- The differential entropy of this distribution is $\log(b - a)$
- What happens if I scale the real numbers, so a to b becomes $c \cdot a$ to $c \cdot b$?
- Differential entropy changes!

This means I am more uncertain about the distribution of heights in millimeters than in centimeters than in inches...

...But Then Again Maybe It Is?

I need more digits to capture the same quantity in mm than I do in cm
than I do in inches than I do in feet

Solution!

Relative Entropy Differences

$$I(X; Y) = H(X) - H(X | Y) = H(Y) - H(Y | X)$$

Properties:

- Exists and is the same natural thing for both discrete and continuous RV
- $I(X; Y) = 0$ if and only if X and Y are independent

$$D_{\text{KL}}(P \parallel Q) = H(P, Q) - H(P) = \sum_x P(x) \log \frac{P(x)}{Q(x)}$$

- Number of extra bits needed to encode via the wrong distribution Q vs. the right distribution P
- $D_{\text{KL}}(P \parallel P) = 0$
- Pseudo distance metric for probability spaces
- **NOT symmetric:** $D_{\text{KL}}(P \parallel Q) \neq D_{\text{KL}}(Q \parallel P)$

Statistics

If you can actually randomly (i.i.d.) sample, you can learn a lot from a small amount of data.

If you can't, you're toast.

Other Important Statistical Points

- ① How to estimate properties of a distribution in an unbiased manner (why is the denominator in the sample variance $n - 1$?)
- ② Hypothesis Testing (False discovery rate correction, permutation tests, Lindley's Paradox)
- ③ Correlation tests of different varieties